Sinusoidal Distribution

A new flexible 4-parameter absolutely-continuous probability distribution

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Abstract

We introduce the *Sinusoidal distribution*, a flexible 4-parameter location-scale-skewness-kurtosis family of distributions, based on Gilbert's Sine distribution (1893). The mathematical properties of this distribution are studied. Examples of fitting the distribution to various statistical objects are considered to demonstrate the distribution's flexibility. Finally, some future avenues are considered for exploration.

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1 Introduction

2 The Distribution

3 Mathematical Properties

4 GHGF

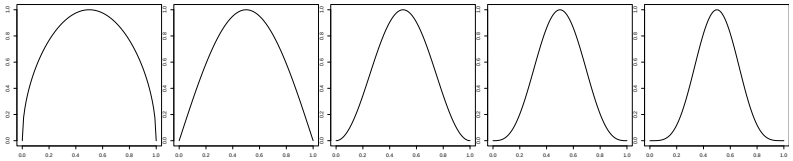
5 Fitting

6 Future Objectives

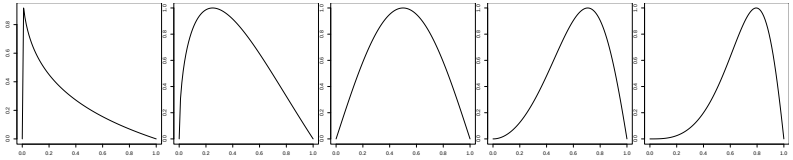
The sinusoidal shape

The sine function over $[0, \pi]$ has a nice 'bell" shape indeed. Let us rescale the argument and deal with $\sin(\pi x)$ over $x \in [0, 1]$.

- But what about its powers – $\sin^k(\pi x)$ as k varies?



- And what about powers of its argument – $\sin(\pi x^s)$ as s varies?



A first attempt at controlling "peakedness" and "inclination" intuitively! We can even combine both the operations to control them simultaneously!

1. Hence the choice $\sin^k(\pi z^s)$ for the kernel!

Other "sine" distributions

Work	Distribution
Sine Distribution, [2]	$f(z) = \frac{\pi}{2}\sin(\pi z)$
Sine-G family, [4]	$F(x) = \sin\left(\frac{\pi}{2}G(x)\right)$
Transmuted Sine-G family, [7]	$F(x) = (1+\lambda)\sin\left[\frac{\pi}{2}H(x)\right] - \lambda\sin^2\left[\frac{\pi}{2}H(x)\right]$
New Sine family, [1]	$F(x) = \frac{\sin\left[\frac{\pi\alpha}{2}G(x)\right]}{\sin\left(\frac{\pi\alpha}{2}\right)}$
Sine generalized-G family, [5]	$F(x) = \sin\left[\frac{\pi}{2}1 - G^{\alpha}(x)\right]$
New Transformed Sine-G family, [6]	$F(x) = \frac{\exp\sin\left[\frac{\pi H(x)}{1+h(x)}\right]-1}{\mathrm{e}-1}$
Sine-Frechet-G family, [3]	$F(x) = \sin\left[\frac{\pi}{2}\exp -b^{a}(-\ln 1 - G(x))^{-a}\right]$

Table: Existing distributions involving the sine function

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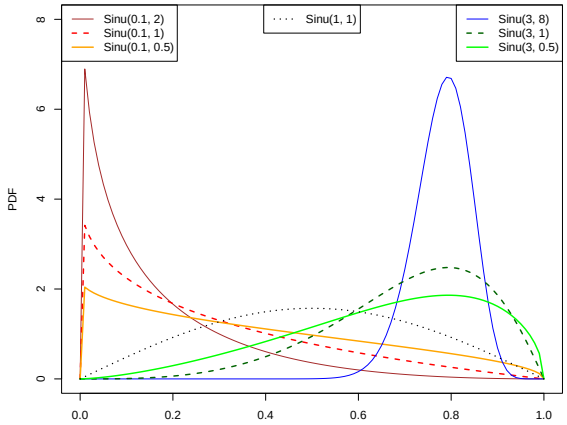
Standard Sinusoidal Distribution

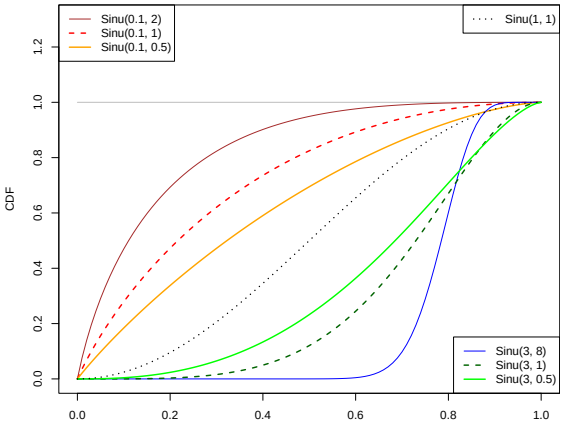
Notation: $Z \sim \mathrm{Sinu}(s, k)$

1. Parameters: $s \geq 0,\ k \geq 0$
2. Kernel: $\varsigma(z; s, k) := \sin^k(\pi z^s)$
3. Normalizing constant $A(s, k) := \int_0^1 \sin^k(\pi t^s)\mathrm{d}t$
4. PDF: $\psi(z; s, k) \equiv \frac{\varsigma(z;s,k)}{A(s,k)} = \frac{1}{A(s,k)} \sin^k(\pi z^s); \quad z \in [0, 1]$
5. $\Psi(z; s, k) = \int_0^z \psi(t; s, k)\mathrm{d}t \equiv \frac{A_z(s,k)}{A(s,k)}; \quad z \in [0, 1]$ where $A_z(s, k)$ denotes the incomplete integral of the kernel

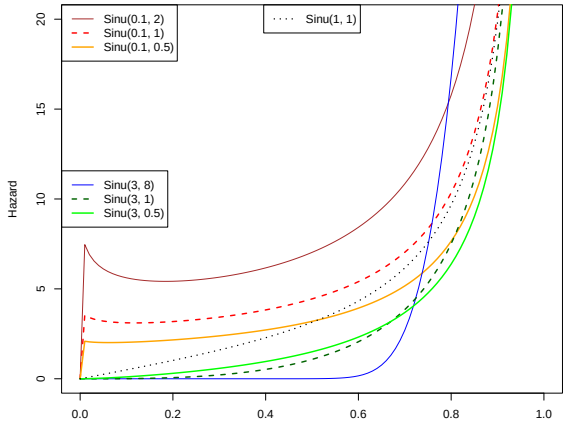
Plots



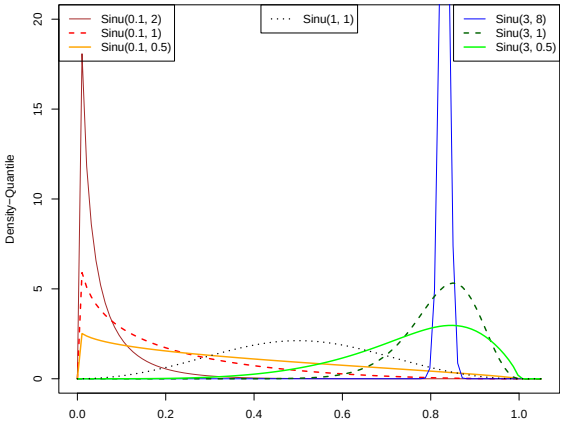
(a) PDFs



(b) CDFs



(c) Hazard Functions



(d) DQFs

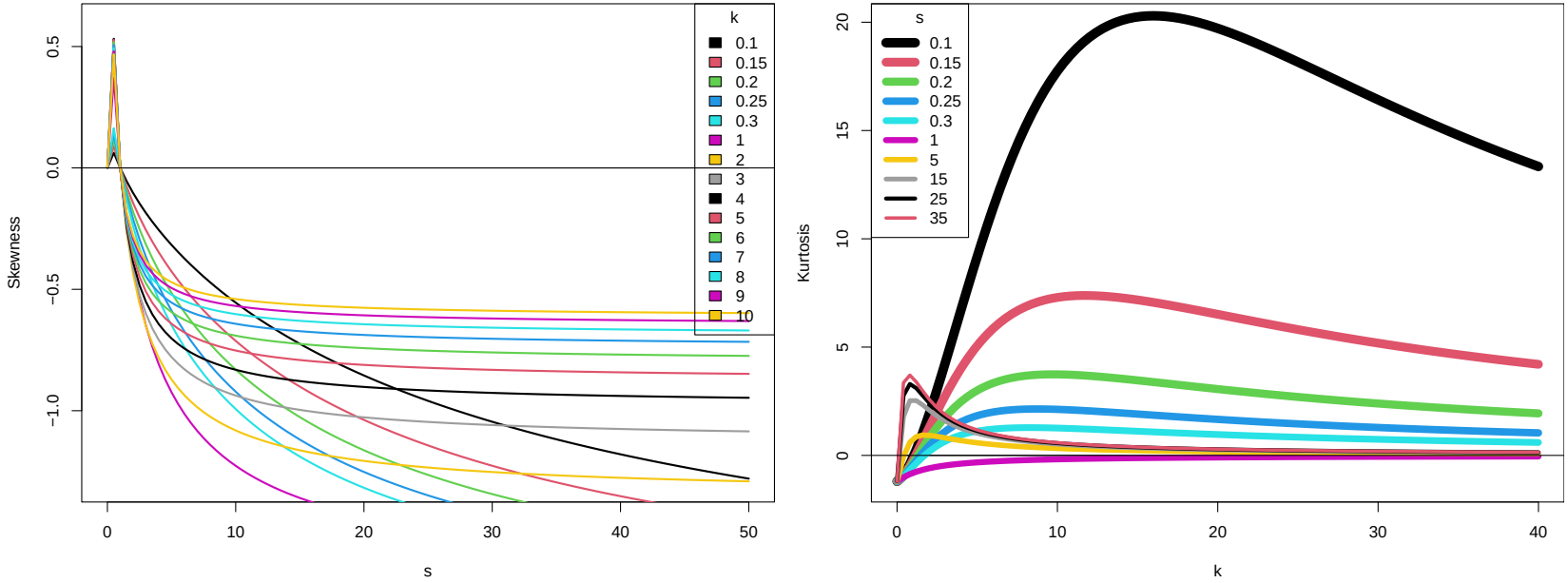
Extensions

1. Location-Scale extension: $\psi(x; a, d, s, k) = \frac{1}{d}\psi\left(\frac{x-a}{d}; s, k\right); \quad x \in [a, a+d]$ with parameters $a \in \mathbb{R}, d \geq 0, s \geq 0, k \geq 0$
2. Flipped extension: $\pi'(x; a, d, s, k) = \frac{1}{d}\psi\left(\frac{a+d-x}{d}; s, k\right)$

Shape properties

1. Parameters: Location: $a \in \mathbb{R}$, $d \geq 0$, $s \geq 0$, $k \geq 0$
2. Support $[a, a + d]$
3. $s = 1$: symmetric. Left and right tilts on respective sides of equality.
4. Increasing k first increases but then decreases kurtosis
5. Mode: $2^{-1/s}$, Modal density: $1/A(s, k)$

Skewness-Kurtosis Interdependence on s, k



(a) Skewness vs. s

(b) Kurtosis vs. k

Figure: s, k jointly affect skewness & kurtosis

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Mathematical Properties I

1. $\mathrm{Sinu}(s,k)$ distribution corresponds to the intensity of crater activity on the planetary surface given by the radial curve $r(z) = r_0 \exp\left(\int_0^{\pi z^s} \tan(t - \sin^{-1} \sin^k t)\mathrm{d}t\right)$
2. $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathrm{Sinu}(a,d,s,k)$ have joint density $f(x_1 \ldots x_n) = \frac{1}{A^n(s,k)2^{nk}d^n}\left(\sum_{r_1=\pm 1}\cdots\sum_{r_n=\pm 1}\left[\prod_{j=1}^n r_j\right]\cdot\exp\left[\sum_{j=0}^n \mathrm{i}\pi r_j\left(\frac{x_j-a}{d}\right)^s\right]\right)^k$
3. $\mathrm{Sinu}(a,d,s,k)$ is sub-Gaussian with parameter $d^2/4$

Mathematical Properties II

4. The normalizing constant for the symmetric case is $A(1,k) = \frac{2^k}{\pi}\frac{\Gamma^2\left(\frac{k+1}{2}\right)}{\Gamma(k+1)} \equiv \frac{1}{\sqrt{\pi}}\frac{\Gamma\left(\frac{k+1}{2}\right)}{\Gamma\left(\frac{k}{2}+1\right)}$
5. Incomplete integral of the symmetric ($s = 1$) kernel for $k \in \mathbb{Z}^+$ is given as:

$$\int_0^z \sin^k \pi t\, dt = \frac{1}{2^{2n}}\binom{2n}{n} z + \frac{(-1)^n}{\pi 2^{2n-1}} \sum_{k=0}^{n-1} (-1)^k \binom{2n}{k} \frac{\sin 2(n-k)\pi z}{2(n-k)} \qquad [k = 2n] \tag{1}$$

$$= \frac{1}{\pi 2^{2n}} (-1)^n \sum_{k=0}^{n} (-1)^k \binom{2n+1}{k} \frac{1 - \cos(2n+1-2k)\pi z}{2n+1-2k} \qquad [k = 2n+1] \tag{2}$$

6. Taking $d = \pi\sqrt{k}$ and letting $k \to \infty$, a symmetric about 0 Sinusoidal density, $\mathrm{Sinu}\left(-\frac{d}{2}, d, 1, k\right)$ tends to a Standard Normal density; i.e. $\psi_k(x) \to \phi(x)$ as $k \uparrow \infty$

Tending normality

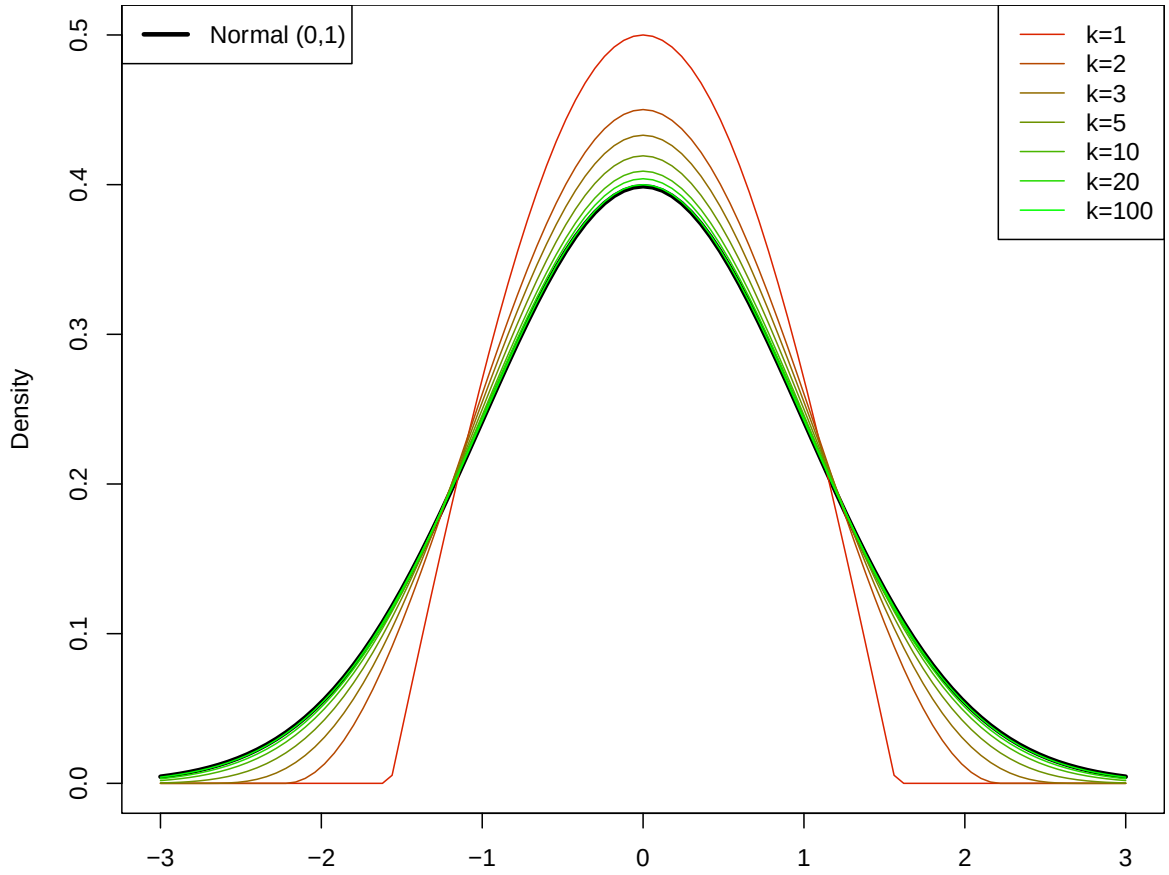

Figure: Sinusoidal distributions tending closer to $\mathcal{N}(0,1)$ as $k \uparrow \infty$

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Context

1. A distribution family without explicit formulae for its CDF, moments, MGF, CF, etc. misses its true potential
2. No matter how flexible, it is difficult to work with on paper
3. Chances are that such a distribution is also a computational nightmare during fitting, inference, etc.

Introducing the GHGF

Generalized Hypergeometric Function (GHGF) F^p_q, parametrized by $a_1, \ldots, a_p$ and $b_1, \ldots, b_p$, calculated at the argument x is:

$$\mathrm{F}^p_q \begin{pmatrix} a_1 & \cdots & a_p \\ b_1 & \cdots & b_q \end{pmatrix}; x \Bigg) := \sum_{j=0}^{\infty} \frac{(a_1)^{\overline{j}} \cdots (a_p)^{\overline{j}}}{(b_1)^{\overline{j}} \cdots (b_q)^{\overline{j}}} \frac{x^j}{j!}$$

where $(x)^{\overline{n}} := x(x+1)\cdots(x+n-1)$ is the **rising factorial / Pochhammer function**.
Also denote $\binom{r}{k} = \frac{(r)^{\overline{k}}}{k!}$ as the **generalized Binomial coefficient**.

Strategies I

Depending on the purpose and the form of the function at hand, the following strategies may work into providing simple explicit expressions for the statistical properties of the *Sinusoidal distribution*:

1. Euler expansion: $\sin\theta = \frac{\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta}}{2\mathrm{i}}$
2. Maclaurin series expansion only when exponents are not involved; otherwise complicated "generalized Multinomial expansion" viz. $\sin^k(x) = \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right)^k$, which quickly becomes intractable.
3. Identifying a GHGF upto a scalar multiple from its ratio of terms, since $\frac{\mathrm{F}_{n+1}}{\mathrm{F}_n} \left[\begin{matrix} a_1 \cdots a_p \\ b_1 \cdots b_1 \end{matrix}; z\right] = \frac{(a_1+n)\cdots(a_p+n)}{(b_1+n)\cdots(b_q+n)} \frac{z}{n+1}$

Properties I

All properties below are described for the Standard *Sinusoidal distribution* $\mathrm{Sinu}(s,k)$. Location-scale families can be easily extended to.

1. The kernel $\sin^k(\pi z^s)$ can be given as $(\pi z^s)^k \left(\mathrm{F}_1^0 \left[\begin{matrix} \cdot \\ \frac{3}{2} \end{matrix}; -\frac{\pi^2 u^2}{4} \right] \right)^k$, or using the GHGF as $\mathrm{F}_0^1 \left[\begin{matrix} k \\ \cdot \end{matrix}; -\exp(-2\mathrm{i}\pi z^s) \right]$.
2. The characteristic function can be given as $\frac{1}{A(s,k)(2\mathrm{i})^k} \sum_{j=0}^{\infty} (-1)^j \binom{k}{j} \sum_{m=0}^{\infty} \frac{(\mathrm{i}t)^m}{(m+1)!} \mathrm{F}_1^1 \left[\begin{matrix} \frac{m+1}{s} \\ \frac{m+1}{s} + 1 \end{matrix}; \mathrm{i}\pi(k-2j) \right]$.

Properties II

3. Thus the MGF becomes
$$\frac{1}{A(s,k)(2\mathrm{i})^k}\sum_{j=0}^{\infty}(-1)^j\binom{k}{j}\sum_{m=0}^{\infty}\frac{t^m}{(m+1)!}\,\mathrm{F}_1^1\left[\begin{matrix}\frac{m+1}{s}\\ \frac{m+1}{s}+1\end{matrix};\ \mathrm{i}\pi(k-2j)\right]$$
4. The expression for the rth order moment is $\mu_r(s,k) = \frac{1}{A(s,k)(2\mathrm{i})^k(r+1)}\sum_{j=0}^{\infty}(-1)^j\binom{k}{j}\,\mathrm{F}_1^1\left[\frac{r+1}{s}\,\frac{r+1}{s}+1;\ \mathrm{i}\pi(k-2j)\right] = \frac{1}{A(s,k)(2\mathrm{i})^k}\sum_{j=0}^{\infty}\binom{k}{j}(-1)^j\sum_{n=0}^{\infty}\frac{[\mathrm{i}(k-2j)\pi]^n}{n!(ns+r+1)}$
5. CDF is given by
$$\Psi(z;s,k) = \frac{1}{A(s,k)(2\mathrm{i})^k}\sum_{j=0}^{\infty}(-1)^j\binom{k}{j}\sum_{n=0}^{\infty}\frac{[\mathrm{i}(k-2j)\pi]^n}{n!(ns+1)}z^{ns+1}$$
6. The normalizing constant is given by
$$\frac{1}{(2\mathrm{i})^k}\sum_{j=0}^{\infty}(-1)^j\binom{k}{j}\sum_{n=0}^{\infty}\frac{[\mathrm{i}(k-2j)\pi]^n}{n!(ns+1)}$$

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Fitting Methods

The following routes for fitting the *Sinusoidal distribution* have been thought of:

1. MLE on sample observations
2. MSM on MSM
3. PDF on PDF
4. CDF on CDF
5. CDF on ECDF

Currently, (3) has been briefly implemented and (1), (2) discussed.

MLE on Data I

Joint likelihood:

$$l(a, d, s, k; \underset{\sim}{x}) = -n \ln(d) - n \ln(A(s, k)) + k \sum_{i=1}^{n} \ln \sin \pi \left(\frac{x_i - a}{d} \right)^s$$

Calculating the preliminary derivatives:

$$\nabla_s A = \pi k \int_0^1 \sin^{k-1}(\pi z^s) \cos(\pi z^s)(\ln z) z^s \, \mathrm{d}z$$

$$\nabla_k A = \int_0^1 \sin^k(\pi z^s) \ln \sin(\pi z^s) \, \mathrm{d}z$$

MLE on Data II

Gives us the following score equations:

$$\nabla_a \ell = 0 \iff \sum_{i=1}^{n} z_i^{s-1} \cot(\pi z_i^s) = 0$$

$$\nabla_d \ell = 0 \iff \pi s k \sum_{i=1}^{n} z_i^s \cot(\pi z_i^s) = -n$$

$$\nabla_s \ell = 0 \iff \pi k \sum_{i=1}^{n} \frac{z^s \cos \pi z^s \ln z}{\sin \pi z^s} = n \frac{\nabla_s A}{A}$$

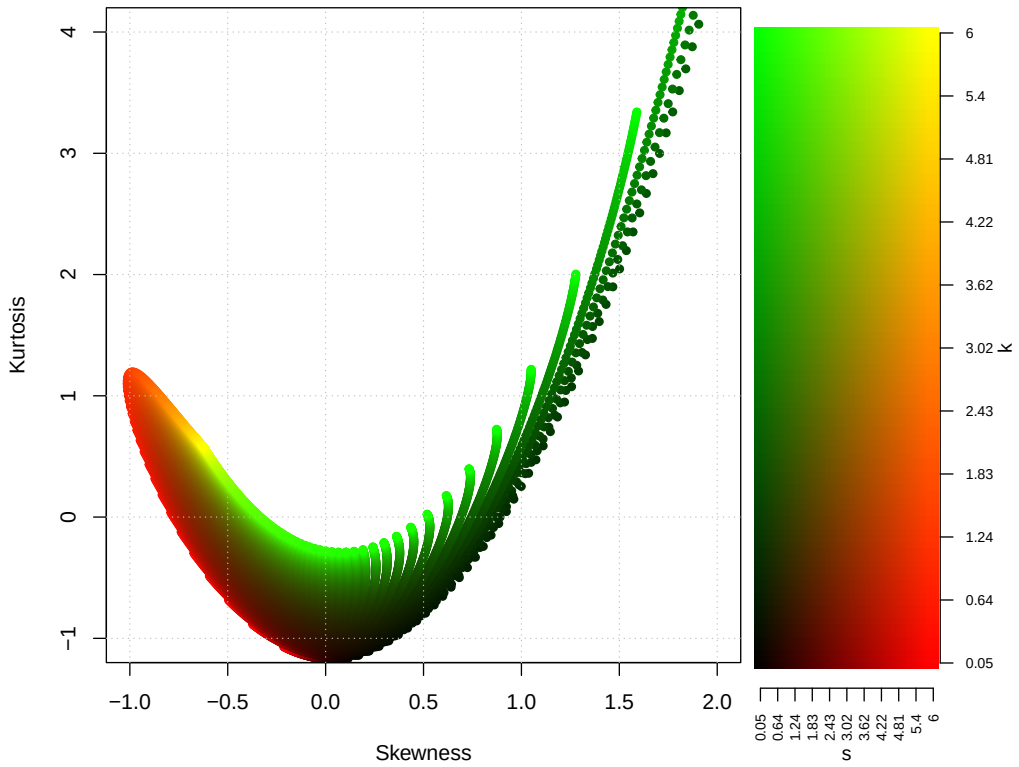
$$\nabla_k \ell = 0 \iff \sum_{i=1}^{n} \ln \sin \pi z_i^s = n \frac{\nabla_k A}{A}$$

MLE on Data III

- Difficult to simultaneously solve
- Computational maximization may be considered

MSM on MSM

The skewness-kurtosis cover-plot of $\mathrm{Sinu}(s, k)$ for the parameter range $s, k \in 0.5\,(0.5)\,6$:



Choice of loss function and optimization methods will be considered in the next iteration.

PDF on PDF I

1. $g(x)$: target density, $f \equiv \psi(x; a, d, s, k)$: candidate density.
2. Hellinger loss function:
   $L(a, d, s, k) = 1 - \int \sqrt{\psi(x; a, d, s, k) g(x)} \mathrm{d}x$
3. Parameter range for computational finiteness: $a \in [-10^3, 10^3]$, $d \in [10^{-2}, 2 \times 10^3]$, $s \in [10^{-2}, 10^2]$ and $k \in [10^{-2}, 10^2]$.
4. `R`'s `optim` function, `L-BFGS-B` method used to optimize the loss function.

PDF on PDF II

Target Density	Hellinger distance (Choice of Parameters a, d, s, k)
$\mathcal{N}(0,1)$	0.00004277289 (-20.542932, 34.985498, 1.304535, 100)
$\text{Lognormal}(0,1)$	0.000936801 (0.02827762, 1000, 0.08566970, 35.78972)
$\text{Skew}\mathcal{N}_{\alpha=1}(0,1)$	2.189446e-05 (-7.2216216, 24.8516728, 0.5936823, 100)
$\text{Logistic}(0,1)$	0.003149183 (-80.275639, 102.222513, 2.882431, 100)
$\text{Exp}(1)$	0.008101119 (0.0005708485, 6.5585684475, 0.0621246472, 1.6766166256)
$\text{DE}(0,1)$	0.01599712 (-8.895071, 16.872288, 1.085768, 15.608238)
$\Gamma(0.1,1)$	Poor fit
$\Gamma(100,1)$	Poor fit
$\text{B}(1,1)$	0.0030202, (-0.0007780335, 1.0012066335, 0.1004892336, 0.12190192670)
$\text{Cauchy}(0,1)$	0.06878475 (-126.211809, 149.303341, 4.141315, 100)
$\chi^2(1)$	Poor fit
$\chi^2(10)$	5.791378e-06 (-0.3201950, 97.1337699, 0.2815837, 22.0271292)
$\text{t}_1(0,1)$	0.06878477 (-126.114092, 149.207078, 4.138119, 100)
$\text{t}_{10}(0,1)$	Fit failed
$\text{F}(1,1)$	Poor fit

PDF on PDF III

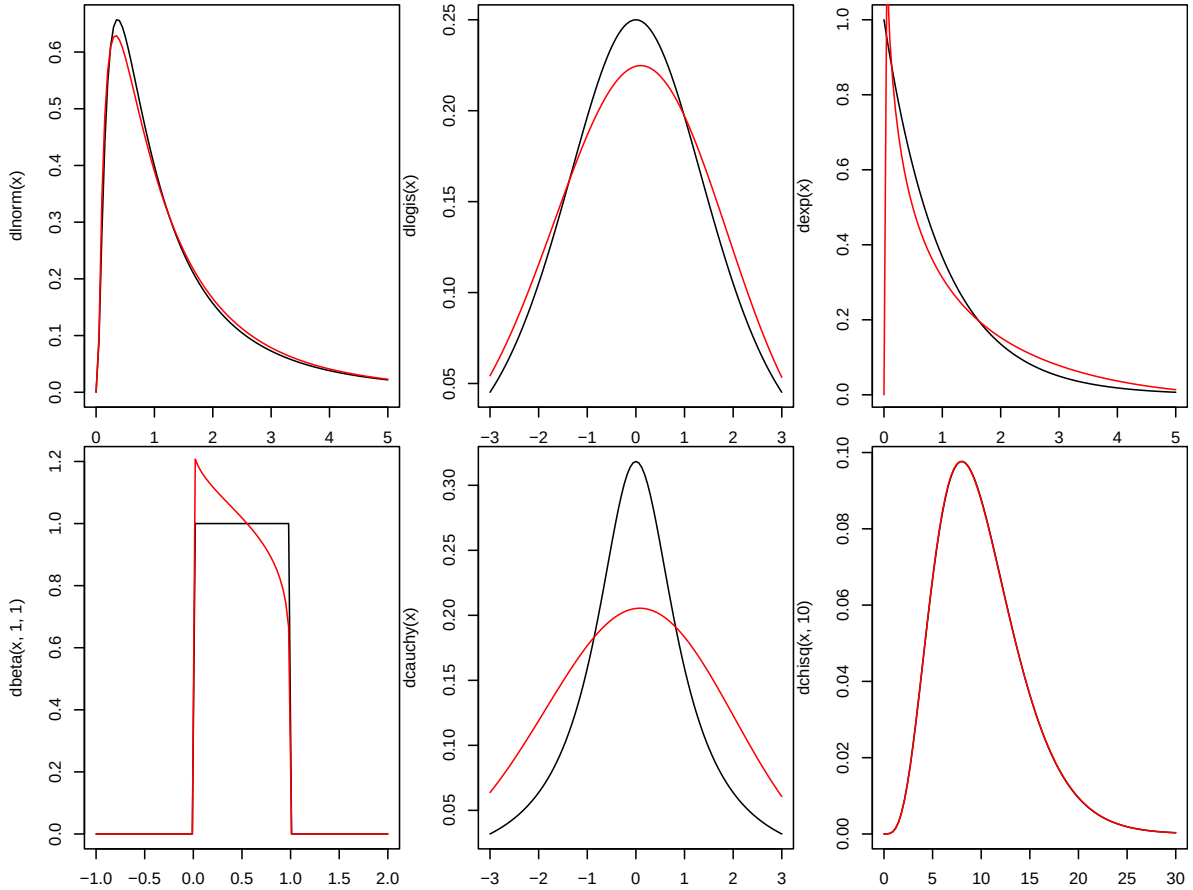


Figure: Some results from PDF on PDF fitting using above methodology

PDF on PDF IV

1. Results greatly dependent on initialization of parameters
2. The nature of the Hellinger metric affects the fit. It will be a task to investigate the effect of other metrics on PDF on PDF fitting.
3. Certain distributions seem to be out of the shape-coverage (at least skewness-kurtosis coverage) of the *Sinusoidal distribution*.

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1. Further investigation and simplification of analytical properties
2. Investigation of its replication capacity, parametric-limiting nature
3. Devising fitting algorithms for more pairs of statistical objects, and newer methods for existing ones
4. Studying the effect of different metrics on fits. Characterization of metrics to some degree
5. Modelling the distribution of complicated statistics
6. Fitting to unconventional datasets, especially in survival statistics and econometrics
7. Usage as flexible Bayesian priors with the further incorporation of variable skewness and kurtosis
8. Usage as a flexible learnable-multiparameter activation function for NNs. Comparison against existing methods.

Thank You

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